Marmara Üniversitesi
Electrical and Electronics Engineering Department
EE 3016: Electromagnetics (Fall 2023)

Instructor Information	Mohammad Alsunaidi	Office M4-232	Office Hours Mon-Thu 11:30-12:30	Phone	Email mohammad.alsunaidi@marmara	
Course Information	**Textbook**: Elements of Electromagnetics, M. Sadiku, 5th Ed.		Grading	Class Work 30%	Mid-term Exam 30%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Exercises
1 Oct 2-6	---	**Vectors.** Review of vector analysis and operations on vector. **Coordinate Systems**. Introduction to rectangular, cylindrical, spherical systems.	1 2	1-8 1-3	**(1)** 3, 5, 21, 22 **(2)** 1, 2, 4, 7
2 Oct 9-13	---	**Coordinate Systems**. Representation of points and vectors in all three coordinate systems and their transformations. **Vector Calculus**. Differentials and integrals.	2 3	3-5 1-2	**(2)** 10, 15, 17 **(3)** 4, 5, 7, 10
3 Oct 16-20	Course drop/add Week	**Vector Calculus**. The del operator: Gradient, divergence, curl, and Laplacian operations.	3	3-8	**(3)** 12, 15, 19, 24, 28, 31, 38, 44
4 Oct 23-27	Cumhuriyet Bayramı Oct 29	Coulomb's Law. Gauss' Law. Application of Gauss' Law	4	1-6	(**4**) 4,5,8,11,16,18,19,20, 22,24
5 Oct 30-Nov 3	Quiz 1 Oct 31 (in-class)	Electric Potential. Electric Dipole, Electrostatic energy density.	4	7-10	(**4**) 27,29,32,35,38,39,41, 42,43
6 Nov 6-10	Assignment 1 Due Nov 10	Electrostatics in materials: Currents. Polarization, Continuity Equation, Boundary Conditions.	5	1-9	(**5**) 3,4,10,12,13,15,19, 20,23,25,28,30
7 Nov 13-17	---	Electrostatic Boundary-value Problems: Poisson's & Laplace's Equations (1D). Resistance & Capacitance calculations.	6	1-5	(**6**) 1,3,6,12,18,20,23, 29,34
Nov 20-26	**Mid-term Exams period**				
9 Nov 27-Dec 1	---	Magnetostatics: Biot-Savart's and Ampere's Laws.	7	1-3	(**7**) 3,7,10,13
10 Dec 4-8	---	Application of Ampere's Law, Magnetic Flux Density, Magnetic Potentials.	7	4-7	(**7**) 19,20,23,25
11 Dec 11-15	Assignment 2 Due Dec 15	Magnetic Forces. Magnetization. Magnetic Boundary Conditions.	8	1, 2, 5, 7	(**8**) 3,5,6,8,9,17,23,26
12 Dec 18-22	Midterm Make-up (Excused, Dec 21-23)	Inductors and Inductance. Magnetic Energy	8	8, 9	(**8**) 28,31,32
13 Dec 25-29	Quiz 2 Dec 26 (in-class)	Electromagnetics: Faraday's Law and its applications.	9	1-3	(**9**) 1,3,4,5,7,11
14 Jan 1-5	New Year 2024 Monday	Displacement Current, Maxwell's Equations, Time-harmonic EM Fields	9	4-5, 7	(**9**) 13,14,15,16,22,28
15 Jan 8-12	Course Project Due Jan 12	**Review**	--	---	---
Jan 15-28	**Final Exams period**	**Jan 29-Feb 4 (Re-sit Final Exams period)**			